

Interval Overlap Check

DESCRIPTION

Two closed intervals on the number line are defined as $[L_1, R_1]$ and $[L_2, R_2]$ (where $L_1 \leq R_1$ and $L_2 \leq R_2$). Using logical conditions, determine whether these two intervals overlap (share at least one common real number point).

INPUT FORMAT

The input contains four integers L_1, R_1, L_2, R_2 , separated by spaces.

OUTPUT FORMAT

Print `overlap` if the intervals share at least one point; otherwise, print `disjoint`. Follow the word with a newline.

SAMPLE

Sample Input	Sample Output
2 8 5 12	overlap